



C23-CM-AI-AIM-CIOT-CCN-303

23190

BOARD DIPLOMA EXAMINATION, (C-23)

MARCH/APRIL—2025

DCME – THIRD SEMESTER EXAMINATION

OPERATING SYSTEMS

Time : 3 Hours]

[Total Marks : 80

PART—A

3×10=30

- Instructions :** (1) Answer **all** questions.
(2) Each question carries **three** marks.
(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. List different types of system call.
2. State the purpose of real time operating system.
3. Write any three differences between program and process.
4. What is FCFS Scheduling algorithm?
5. What is mutual exclusion?
6. List necessary conditions for arising deadlocks.
7. Define the term dynamic loading.
8. What is single partition memory allocation?

9. List various file operations.
10. Define the terms seek time and transfer time.

PART—B

10×5=50

- Instructions :** (1) Answer *any five* questions.
(2) Each question carries **ten** marks.
(3) Answers should be comprehensive and the criterion for valuation is the content but not the length of the answer.

11. Explain single user operation system structure and multiuser OS Structure.
12. For the processes given in table, draw a Gantt chart illustrating their execution using priority scheduling (both preemptive and non-preemptive). A higher priority number has higher priority. Find out average waiting time of each process.

Priority	Process no.	Arrival Time	Burst Time
4	1	0	10
5	2	1	5
8	3	2	8
12	4	3	4
9	5	5	3
10	6	6	6

13. Explain about interprocess communication.

14. A system has 4 processes and 5 allocated resources. The current allocation and maximum needs are as follows :

	Allocated					Maximum				
A	1	0	2	1	1	1	1	2	1	3
B	2	0	1	1	0	2	2	2	1	0
C	1	1	0	1	1	2	1	3	1	1
D	1	1	1	1	0	1	1	2	2	0

If Available = [00 X 11], what is the smallest value of X for which this is a safe state?

15. Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 300 KB and 250 KB. These partitions need to be allocated to four processes of size 357 KB, 210 KB, 468 KB and 491 KB in that order. Perform the allocation of processes using first-fit allocation method?
16. Explain how logical address is translated into physical address.
17. Consider a disk queue with requests for I/O to blocks on cylinders 98, 183, 41, 122, 14, 124, 65, 67. The FIFO scheduling algorithm is used. The head is initially at cylinder number 53. The cylinders are numbered from 0 to 199. What is the total head movement (in number of cylinders) incurred while servicing these requests?
18. Explain various file access methods in detail.

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